Sean Kakhkhorov

AI Engineer · RAG & Retrieval · Backend · DevOps

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SUMMARY

AI engineer and full-stack developer, **2 years 10 months of professional experience**. I build RAG assistants end to end: hybrid retrieval, evaluation, structured output, quote checks, APIs and deployment. **Five live products** with LLM features, four solo; CargoCheck AI turns the invoice checks from my International Trade degree into software.

SKILLS

LLM	Gemini API (structured output, embeddings), RAG, quote checks, abstention, Zod / Pydantic schemas
Retrieval	BM25, pgvector, Chroma, Postgres full-text search, RRF fusion, Kiwi Korean tokenizer, article chunking
Evaluation	hit@k, recall@k, MRR on labelled question sets · pytest, Jest, Vitest, Supertest
Backend	Python, FastAPI, TypeScript, Node.js, NestJS, GraphQL, REST, BullMQ, OCR (Tesseract)
Data	PostgreSQL (Neon), Drizzle ORM, MongoDB, Redis (Upstash)
DevOps	Docker, Google Cloud Run, GitHub Actions, Vercel, Render, Nginx, Linux · React, Next.js

EXPERIENCE

IT Local — Full-stack Web Developer

Uzbekistan · 2023.11 – 2026.02

- Built websites and APIs for **20+ client companies** in a team of 4+ developers (React, Node.js APIs, auth).
- Set up **5–15 Ubuntu servers** (Nginx, PM2/Docker) and replaced manual SSH deploys with a **CI pipeline**.

Atlantis Investment — Backend Developer

Uzbekistan, alongside studies · 2023.04 – 2023.09

- Built Express.js REST APIs with JWT/session sign-in, uploads, and MongoDB indexes for search and paging.

PROJECTS

Korea Visa Assistant

Live · beta | Solo · bilingual RAG | Live app | GitHub

- EN/KO RAG over 44 statute chunks (21 visa categories): article chunking, Kiwi tokens, BM25 + Chroma, RRF (k = 60).
- **32 synthetic source-labelled questions**: hybrid **hit@5 0.969, recall@5 0.891, MRR 0.935** (BM25 0.702); retrieval only.
- Quote-checked Gemini claims, abstention, identifier rejection; pinned index snapshots on Cloud Run; **125 pytest** + 5 JS tests.

Sofiya

Live | Solo · Ask Sofiya RAG | sofiya.store | GitHub

- pgvector + full-text search fused with RRF; Gemini recommends only in-stock books, with verified citations.
- **45 questions** incl. paraphrased and unanswerable: **hit@5 100%, hit@1 93%, MRR 0.95** vs keyword-only hit@5 85%.
- Next.js 15, NestJS 11, Neon; Cloud Run with keyless CI/CD; Redis daily AI cap; **~260 automated tests**.

CargoCheck AI

Live | Solo · document extraction | Live app | GitHub

- Gemini structured extraction (Zod) in BullMQ jobs; people confirm values before quantity, unit and decimal checks.
- Per-flag decisions, PDF reports; Vercel, Render (Docker), Neon, Upstash; **42 automated tests**.

CreatorHub

Live | Solo · 2026.07 – present | creatorhub.site | GitHub

- Creator marketplace (EN/KO/UZ) with a Gemini site assistant; Next.js, NestJS GraphQL, MongoDB, Socket.IO.
- Duplicate-safe request keys, restart-safe worker, Docker/Nginx, health-checked rollback; **293 automated checks** (Oct 2026).

Lexara

Live · collaborative | Contributor · backend | Live site

- Worked on the FastAPI backend of a cited document Q&A (RAG) platform: endpoints, uploads, models, migrations.

EDUCATION

Chonnam National University — B.A. International Trade, taught in Korean · 2019.09 – 2023.08

Languages: **Korean** business (4-year degree taught in Korean) · **English** fluent · **Uzbek** native · **Russian** elementary